MIL-A-21071B(AS)
2 March 1973
SUPERSEDING
MIL-A-21071A(WP)
16 March 1965

MILITARY SPECIFICATION

ANTIFOGGING COMPOUND, TRANSPARENT AIRCRAFT ENCLOSURES

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

* 1.1 Scope - This specification establishes the formulation and requirements for an antifogging material that is dissolved in a fluorocarbon type solvent.

* 1.2 Classification - The antifogging compound shall be furnished in the following types as specified.

Type I - Solution
Type II - Aerosol propelled

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-F-1421	Fluorocarbon Refrigerants
QQ-A-250/5	Aluminum Alloy Alclad 2024, Plate and Sheet
QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-N-95	Naphtha, Aliphatic
PPP-C-96	Can, Metal, 28 Gage and Lighter

FSC 6850

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:38Z
Check the source to verify that this is the current version before use.

Military

MIL-P-5425	Plastic, Sheet, Acrylic, Heat Resistant
MIL-S-7952	Steel, Sheet and Strip, Uncoated, Carbon (1020 and 1025) (Aircraft Quality)

STANDARDS

Federal

FED-STD-791	Lubricants, Liquid Fuels and Related Products; Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-290	Packaging, Packing and Marking of Petroleum and Related Products

(Copies of specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials - The manufacturer shall certify by affidavit that the antifogging compound conforms to the requirement and proportions specified in Table I. (See 4.4.1)

* 3.1.1 Preparation of compound - The polyoxyethylene (4) sorbitan monostearate shall be gently heated and melted on a hot plate. (CAUTION: This material will be decomposed by excessive heating above the melting point.) The melted polyoxyethylene (4) sorbitan monostearate shall be added slowly to the trichlorotrifluoroethane and dissolved. Then the polyoxyethylene (20) sorbitan monooleate, the coupling agent (hexylene glycol) and the salt-free amphoteric detergent shall be added. The mixture shall be thoroughly agitated until the solution is uniform in composition.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:38Z
Check the source to verify that this is the current version before use.

* TABLE I. REQUIREMENTS AND PROPORTIONS

Ingredient	Percent (By Volume)
Polyoxyethylene (4) Sorbitan Monostearate 1/	5.0 ± 1
Polyoxyethylene (20) Sorbitan Monooleate 2/	12.0 ± 1
Trichlorotrifluoroethane (Type 113, BB-F-1421)	Quantity to make 100 percent
Salt-free Amphoteric Detergent 3/	0.2 ± 0.05
Hexylene Glycol	8.0 ± 1

1/ Shall have a hydroxyl number that is not less than 170 nor more than 200.

2/ Shall have a hydroxyl number that is not less than 65 nor more than 80.

3/ Mineral CAM-SF (Miranol Chemical Company) or equivalent

* 3.2 Aerosol composition (Type II) - The composition of the contents of the aerosol container shall be as in Table II.

* TABLE II. AEROSOL COMPOSITION

Ingredient	Percent (By Volume)
Formulation of Table I	1.5
Trichloromonofluoromethane (Type 11, BB-F-1421)	82.0
Dichlorodifluoromethane (Type 12, BB-F-1421)	16.5

3.3 Effect on acrylic plastics - When tested as specified in 4.5.1, the compound shall cause no crazing, chemical attack or any other deterioration of the acrylic plastic strips.

* 3.4 Flash point - When tested by the open cup method specified in 4.5.2, the flash point of the compound shall be not less than 140°F (60°C).

3.5 Corrosiveness - After immersion in the compound for 4 hours as specified in 4.5.3, the steel, cadmium plated steel and clad aluminum alloy test panels shall show no sign of discoloration, corrosion, chemical attack or any other deterioration; except that a slight darkening of the steel panel shall be acceptable.

Source: https://assist.dla.mil -- Downloaded: 20260-19T08:38Z
Check the source to verify that this is the current version before use.

3.6 Workmanship - When examined visually, the compound shall be a homogeneous mixture that is free from lumps and foreign matter. The odor shall not be obnoxious.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility - Unless otherwise specified by the Government, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the examinations and tests set forth in the specification where such are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot -

* 4.2.1 Type I compound - Each inspection lot shall consist of all compound produced from the same ingredients, offered for acceptance at one time.

* 4.2.2 Type II compound - An inspection lot shall consist of all the aerosol containers, filled under basically the same conditions with compound produced from the same ingredients, offered for acceptance at one time.

4.3 Lot acceptance samples -

* 4.3.1 Samples for tests - A quantity of filled unit containers shall be selected at random from each lot in accordance with MIL-STD-105 at inspection level S-1. Sample quantity for test shall be one pint of Type I compound or two 1-1/2 ounce aerosol containers (Type II).

4.3.2 Sampling of Preparation for Delivery - A quantity of filled unit containers and shipping containers fully prepared for delivery shall be taken at random from each lot in accordance with MIL-STD-105 at inspection level S-1.

4.4 Lot acceptance inspections -

4.4.1 Inspection of bulk material - To avoid rejection after an extensive filling operation, the manufacturer shall examine and test a sample from each lot or batch before filling to insure that the lot or batch conforms to applicable requirements of this specification.

4.4.2 Inspection of samples for tests - Each sample selected in accordance with 4.3.1 shall be examined and tested to determine conformance to all the requirements of Section 3 of this specification except 3.1 and 3.2. The manufacturer

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:38Z
Check the source to verify that this is the current version before use.

shall certify by affidavit that the material to be supplied conforms to the requirements set forth in 3.1 and 3.2, if applicable. Non-conformance of a sample to a requirement shall be cause for rejection of the lot represented by the sample.

4.4.3 Inspection of Preparation for Delivery - Each sample of a filled unit container or shipping container fully prepared for delivery selected in accordance with 4.3.2 shall be visually examined for defects of the containers and closures, for evidence of leakage, for unsatisfactory marking, for net contents and to determine compliance with all the other requirements of Section 5 of this specification. If the number of defective units of product in the lot sample exceeds the acceptable quality level (AQL) of 2.5 percent, the lot represented by the sample shall be rejected.

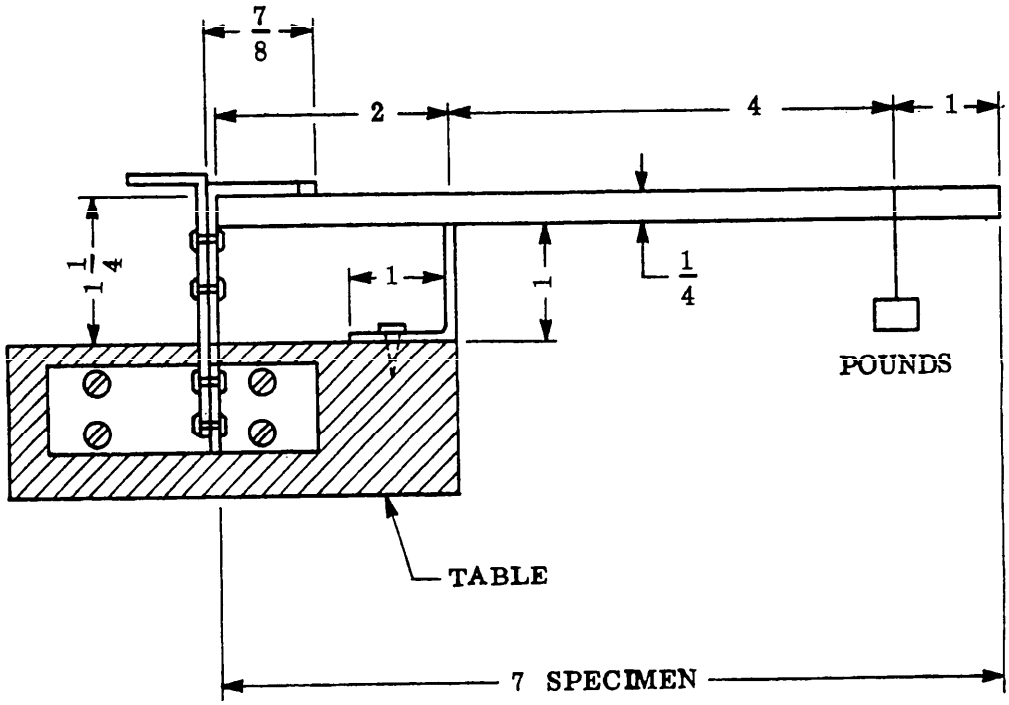
4.5 Test methods - Unless otherwise specified herein, all tests shall be performed at a laboratory room temperature of 72 ± 5°F (22.2 ± 2.8°C).

4.5.1 Effect on acrylic plastics -

4.5.1.1 Test specimen - Four adjoining plastic strips, 1 by 7 by 1/4 inches, shall be cut from acrylic plastic conforming to MIL-P-5425. The plastic shall be cleaned with aliphatic naphtha conforming to TT-N-95 and then allowed to air dry at laboratory room temperature for a minimum of 96 hours, prior to use in test.

4.5.1.2 Procedure - The testing equipment shall be arranged in accordance with Figure 1. The four plastic strips shall be set up as class 1 levers with the fulcrum 2 inches from the clamped end and load suspended four inches from the fulcrum. The strips shall be loaded to produce an outer fiber stress of 2000 pounds per square inch (psi) at the fulcrum. Thirty minutes after the strips have been stressed, they shall be examined for crazing. The examination shall be made over a 7-power magnifier while the strips are under stress. Place a light source under the strips and observe the fulcrum area from above at such an angle that light will be reflected to the eye from the surfaces of any cracks or fissures which are present. To prevent overheating of the plastic strips, the light source shall be utilized only during the examination for crazing. Strips showing craze shall be replaced with others that do not craze after 30 minutes stress. Anti-fogging compound shall then be applied at the fulcrums of three plastic strips (the fourth is a control) by lightly passing a flannel cloth saturated with the compound across the width of the strip and covering an area 1 inch on each side of the fulcrum. Re-apply the compound in the same manner periodically each hour until a total of seven applications have been made. Twenty-four hours after the first application of compound and after the strips have been under stress for 24 hours, examine each strip for crazing as described above. All cracks at the edges shall be noted as "edge crazing" and disregarded unless they extend across the entire width of the strip. If the control crazes, repeat the test using four different test specimens. Examine also for signs of chemical attack or any other deterioration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:38Z
Check the source to verify that this is the current version before use.



NOTES:

1. DIMENSIONS IN INCHES.
2. 1/8 ANGLE IRON USED AT FULCRUM AND END OF SPECIMEN.

Figure 1. Test Set-up For Crazing Resistance of Plastics Under Stress.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:38Z
Check the source to verify that this is the current version before use.

4.5.1.3 Calculation - The load to be applied to each test strip shall be calculated as follows:

$$P = \frac{SBD^2}{24}$$

P = Load (in pounds) applied at the free end of strip, 4 inches from the fulcrum.
B = Width of the strip in inches 1/
D = Thickness of the strip in inches 1/
S = 2000 = stress in pounds per square inch.

1/ Make measurements to nearest 0.001 inch.

4.5.2 Flash point - The flash point shall be determined in accordance with Method 1103.6 of FED-STD-791.

4.5.3 Corrosiveness - The test panels, 1 by 1 by 0.050 inches, shall be thoroughly cleaned with wet pumice, rinsed with ethyl alcohol and dried before use. Two panels shall be prepared from steel conforming to MIL-S-7952, one of which is cadmium plated in accordance with QQ-P-416; and one panel from aluminum alloy (Alclad 2024) conforming to QQ-A-250/5. Place the panels in a Petri dish and cover completely with the antifogging compound. After 4 hours immersion, rinse the panels thoroughly with distilled water and allow to air dry. Examine each panel for signs of discoloration, corrosion, chemical attack or any other deterioration.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging, packing, and marking - Packaging, packing and marking shall be in accordance with MIL-STD-290, except for level A packaging for Type II compound (5.1.1), and additional markings (5.1.2). Unless otherwise specified, the unit container for Type I Compound shall be an oblong, pint can with a metal screw cap closure conforming to MIL-STD-290.

* 5.1.1 Level A packaging - Type II compound shall be packaged in pressurized containers which shall hold 1-1/2 ounces. The containers shall conform to Type IX, Class 2 of PPP-C-96 with a valve opening diameter suitable for the specified valve.

5.1.2 Additional marking - Unit containers shall be marked with the following:

a. Manufacturer's directions for use
b. Lot number
c. CAUTION: USE WITH ADEQUATE VENTILATION. DO NOT EXPOSE LIQUID COMPOUND OR ITS VAPORS TO OPEN FLAME.

Source: https://assist.dla.mil -- Downloaded: 20260-19T08:38Z
Check the source to verify that this is the current version before use.

* 5.1.2.1 In addition to the markings specified in 5.1.2, the unit containers for the Type II compound shall be marked with the following information:

WARNING

PRESSURIZED CONTAINER
Do not store at temperatures above 120°F (49°C). Keep away from direct sunlight, radiators, hot water or other heat sources. Do not puncture or incinerate this can. Use within 12 months of date of manufacture.

6. NOTES

6.1 Intended use - Use of the antifogging compound is intended to prevent the accumulation of fog and haze on the interior surfaces of aircraft windshields and canopies during adverse weather conditions. The compound is suitable for application on either glass or plastic surfaces.

6.2 Ordering data -

a. Title, number and date of this specification
b. Type of compound
c. Size of container and quantity
d. Levels of packaging and packing (see 5.1)

6.3 Marginal notations - The outside margins of this specification have been marked to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Preparing Activity
Navy - AS
Project No. 6850-N476

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-A-21071B(AS) ANTIFOGGING COMPOUND, TRANSPARENT AIRCRAFT ENCLOSURES

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:38Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, PA 19112

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE NAVY
DOD 316



Engineering Specifications and Standards Department (Code X)
Naval Air Engineering Center
Philadelphia, PA 19112

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:38Z
Check the source to verify that this is the current version before use.